Fiscal Year:	FY 2004	**Task Last Updated:**	FY 09/14/2004
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	The Interaction Of Touch And Gravitropic Signaling In Plant Roots		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	10/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):			
Grant/Contract No.:	NAG2-1594		
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Performance Goal Text:			
Task Description:	Touch and gravity are two of the many stimuli that plants must integrate to generate an appropriate growth response. Due to the mechanical nature of both of these signals, shared signal transduction elements could well form the basis of the cross-talk between these two sensory systems. Two of the most widely proposed candidate signal transduction elements in both plant touch and gravity sensing are changes in Ca2+ and pH in the cytoplasm of the sensory cells. However, the precise spatial and temporal characteristics of changes in these signaling elements in response to mechanical and/or gravity stimulation, and especially how they might interact in a signaling cross-talk system, are poorly defined. The overall goal of this experimental program is therefore to seek to define the spatial and temporal character of the ionic signaling associated with mechano-perception. Using this characterization we will then mimic or block the appropriate dynamics of these ionic changes and so define how they regulate the signaling systems responsible		

	for the gravitropic response and generation of plant form. Touch stimulation has dramatic effects on plant morphogenesis and its interaction with gravimorphogenesis has a critical role in determining overall plant development and productivity. In addition, it may be possible to manipulate the gravity response system of the plant with appropriate mechanical stimulation to circumvent problems of plant growth in the weightless environment of space flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is employing state-of-the-art cell biological tools to elucidate the molecular and cellular mechanisms that translate the force of gravity into oriented growth of the plant. In addition, the research aims to define how other signals perceived by the cell might impact on this regulatory system and the molecular mechanisms whereby plants exhibit touch sensitivity. An improved understanding of the regulation of plant gravity sensing mechanisms and how these are altered by the plant's environment should help design approaches to enhance plant growth in microgravity, providing increased plant productivity during sustained space flight and increased control over future spaceflight experiments.
Task Progress:	This research has defined the spatial and temporal characteristics of the signaling systems used by the plant root to sense and respond to gravity and mechanical signaling. Gravity sensing involves a pH signal in the gravity sensing cells of the root cap that is required for full gravitropic response. Thus, chemically blocking this change blocks the plant growth response to gravity. Conversely, in mutants disrupted in the gravity sensing machinery such as the arg1 mutant, the pH change is disrupted. This pH increase acts as an amplifier of the initial gravity signal, likely acting to enhance the transport processes for the hormones that coordinate plant growth in response to gravity stimulation. Touch stimulation induces a transient calcium increase in the root cells. This calcium increase moves from cell to cell throughout the root cap through a calcium-induced-calcium release system, most likely associated with vacuolar calcium release channels. The touch system that generates the initial calcium increase is dependent on the microtubule cytoskeleton whereas cell to cell propagation of this calcium wave is not. Gravity sensing appears to operate through the actin-based cytoskeleton. Disrupting the actin causes a prolonged gravity pH signaling response that leads to an overshoot in the gravity growth response. These results suggest actin is acting to attenuate the gravity signaling system. The calcium increase induced by touch interacts with the pH signaling system by inhibiting this response. Artificially mimicking the calcium increase using photoactivation of a caged calcium ionophore leads to inhibited gravity response even in the absence of a touch stimulus. Thus, touch and gravity signaling appear to interact through their ionic signaling systems. We have screened a T-DNA insertional mutant library for plants lacking a touch response in their roots. Of 10,000 lines screened, 2 putative mutants have been identified and these are currently being genetically characterized and cloned.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Hou G, Kramer VL, Wang YS, Chen R, Perbal G, Gilroy S, Blancaflor EB. "The promotion of gravitropism in Arabidopsis roots upon actin disruption is coupled with the extended alkalinization of the columella cytoplasm and a persistent lateral auxin gradient." The Plant Journal. 2004 Jul;39(1):113-25. PubMed PMID: 15200646 , Jul-2004
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Dissertations and Theses	Massa GD. "The interaction of touch and gravity in plant roots." Ph.D. Thesis, The Pennsylvania State University, August 2003. , Jan-2004
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Presentation	Gilroy, S. "Integrating tropisms: ionic signaling mediates the interaction of touch and gravity in the root cap." Biology Department, University of Missouri, Columbia. Oct-2003
Presentation	Gilroy, S. "Getting to the root of tropisms: ionic signaling and root growth." Biology Department, Stanford University. Nov-2003
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Presentation	Swanson, S.; Gilroy, S. "Role of calcium transporters in the touch and gravity responses of Arabidopsis roots." 15th Penn State Symposium in Plant Biology. May-2004
Presentation	Gilroy, S. "Getting to the root of tropisms." John Innes Center, Norwich, UK. Jan-2004
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Presentation	Blancaflor, E.B.; Hou, G.; Kramer, V.L.; Wang, Y.-S.; Chen, R.; Gilroy, S. "The actin cytoskeleton is a negative regulator of root gravitropism." Annual meeting of the American Society for Plant Biology. Jul-2004

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Presentation	Swanson, S.J.; Gilroy, S. "Role of calcium transporters in the touch and gravity responses of Arabidopsis roots." Annual meeting of the American Society for Plant Biology. Jul-2004
Presentation	Gilroy, S. "Getting to the root of tropisms." PennState University Plant Physiology Program. Sep-2004
Presentation	Gilroy, S. "Integrating tropisms in the root." International Society for Plant Growth Substances. Sep-2004